

Chance Miller

Intro to Electrical Physics and Components

Electrical Charge

- Electrons have negative charge, protons have positive charge
- A circuit moves charges around
- Regions w/ extra electrons have negative charge

Voltage: Electrical Potential energy

- Amount of work needed to keep opposite charges apart
- Ground is a common point
- Measured across 2 points

Current: Flow of Charge

- Measured through a conductor
- Current in = Current out

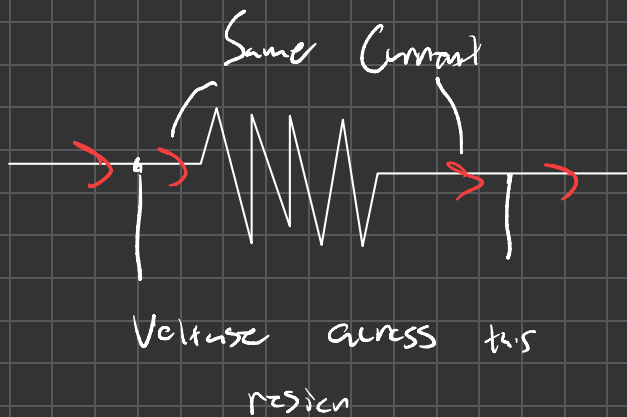
Resistance: How much a material resists current

- Property of the material

Ohm's law:

$$V = IR$$

Voltage Resistance
Current



Power: Instantaneous Rate energy is transferred

$$P = IV$$

• Substitutions using Ohm's law yields

$$P = I^2 R$$

$$P = V^2 / R$$

• Rate energy is transferred to heat

Energy:

- The ability to make a change to work
- Can be stored and transferred from one form to another

Capacitance: How well an object can store charge

- Imagine a reservoir with a source in and source out. A big well takes longer to fill compared to a small one. The size of the well is analogous to capacitance

Inductance: How much an object resists a change in current through it

- Imagine a heavy turbine in a pipe of water. Inductance is the moment of inertia for the turbine. Once the turbine spins, it tries to keep the same current

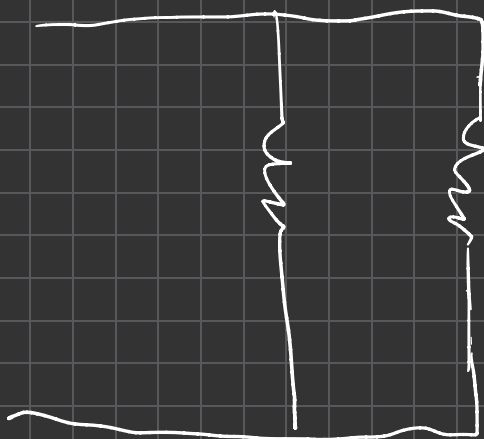
Circuits:

- A loop that allows current to flow to where it comes from
- Made of nodes and devices between those nodes

Series and Parallel



Series



Parallel

- Current is the same between devices

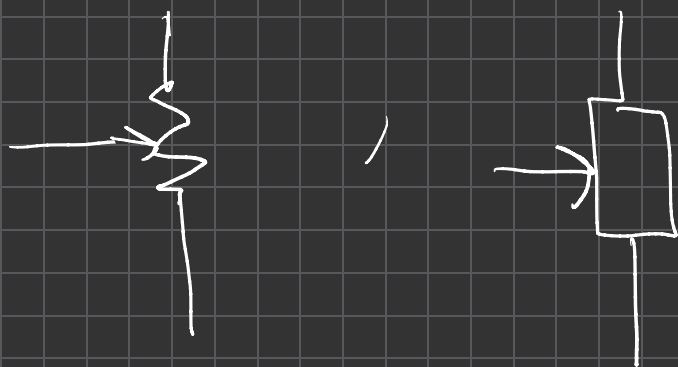
Components:

Resistor: Component w/ set resistance



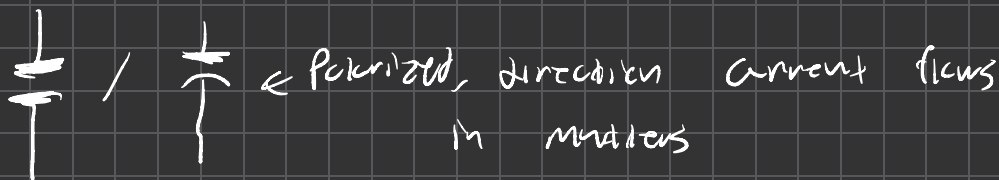
Thermistor: Change resistance depending on temperature

Potentiometer: Resistor w/ variable resistance



Capacitor: Component designed to contain a specific capacitance

• Have memory of past voltage



Inductors: Component designed to contain or
specific inductance

- Has "memory" of past current



Wires: Carries current

- More current = more heat
- Narrow wires $\rightarrow$ higher resistance per length $\rightarrow$ more heat
- Thinner wires have more resistance and more length to dissipate the increased heat

Fuses: protect "downstream" circuits

- Thin wires that blow up before fuses they protect

Relays: Electrical switches isolated from another circuit



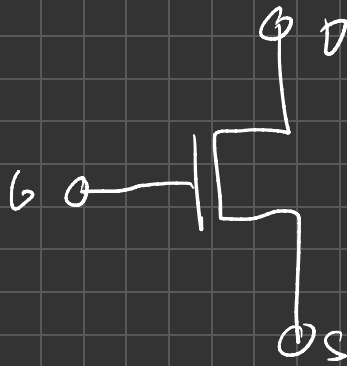
It can be used to
separate high/low
voltage systems

Diodes: Device that restricts current to flow in one direction



Transistor: Switches with no moving parts (unlike relays)

• NMOS:

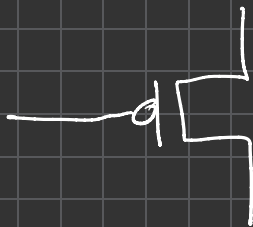


When G is low, current doesn't flow from

D to S . Current flows from D to S

When G is high. G has to be above S

• PMOS:

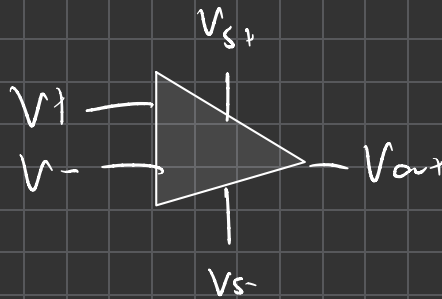


Opposite of NMOS,

current flows when

G is low

OP Amps: Amplifies really small circuits or buffers them to protect fragile signals



Comparator: Gives a digital output depending on analogue input

Linear Regulator: Steps a voltage down to a fixed (sometimes adjustable) voltage